Formal Language & Automata Theory_2024_PYQ

Group-A (Very Short Answer Type Question)

Answer any ten of the following: [1 x 10 = 10]

1. What type of grammar is Context-sensitive grammar?
2. Give two applications of TM.
3. What is a universal Turing machine?
4. Give two applications of Context free languages.
5. Give two examples/applications designed as finite state systems.
6. What is context-free language?
7. Define context sensitive grammar.
8. How can you proceed for Turing machine construction?
9. What are the various representations of TM?
10. Define finite automaton.
11. What do you mean by k-equivalent states?
12. When a grammar is said to be ambiguous?

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Group-B (Short Answer Type Question)

Answer any three of the following: [5 x 3 = 15]

2. Find context-free grammars for the language
(with $n \geq 0, m \geq 0, k \geq 0$): $L = a^n b^m c^k, k = n + 2m.$ [5]
3. Discuss Universal Turing Machine (TM). [5]
4. Show that the following grammar is ambiguous: $S \rightarrow aSbS \mid bSaS \mid \lambda$. [5]
5. Consider the grammar $G = (S, a, b, S, P)$ with P given by $S \rightarrow aSb;\ S \rightarrow \varepsilon$. What will be the language of the grammar? [5]
6. What is the Turing Machine Halting Problem? [5]

Group-C (Long Answer Type Question)

Answer any three of the following: [15 x 3 = 45]

7. Discuss with examples how to simplify context-free grammar:

a) By removing the useless production;

b) By moving λ-production;

c) By removing unit production. [5+5+5]

8. (a) What is Turing Machine? [5]

(b) How can Turing Machines be used as language acceptors? [3]

(c) For Σ = {0,1}, design a Turing machine that accepts the language denoted by the regular expression 00*. [7]

9. (a) When a Problem is said to be Undecidable? [5]

(b) Let $\Sigma = a, b$. Write down the grammar that generates the following language:

$L = a^n b^m : n \geq 0, m < n.$ [5]

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(c) Show that the grammars $S \rightarrow aSb|bSa|SS|a$ and $S \rightarrow aSb|bSa|a$ are not equivalent. [5]

10. (a) Consider the language$L = a^n : n = 3\ or\ n$ is even. Construct a DFA for this language.[5]

(b) For $\Sigma = a, b$, construct a DFA that accepts all strings with an even number of a's. [5]

(c) For $\Sigma = a, b$, construct DFA that accepts all strings with exactly one a. [5]

11. Discuss Chomsky hierarchy of languages with examples. [15]